

Drivechains: BIP 300, BIP 301, and eCash

Agenda

- The two-way peg problem

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- BIP 300 hashrate escrows

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- BIP 300 hashrate escrows
- BIP 301 blind merged mining
- Paul Sztorc's planned eCash fork

Drivechains

Three separate mechanisms

BIP 300

Peg state, deposits, and
miner-approved withdrawals

BIP 301

A fee market for committing
sidechain blocks

eCash

A separate L1 fork that plans
to enable both

They compose, but they solve different problems.

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- Current implementations can enforce them outside Bitcoin Core through CUSF

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- **Blind merged mining:** buying an L1 commitment to an L2 block without asking the L1 miner to validate that L2

A two-way peg



Deposits move value right. Approved withdrawal bundles move value left.

Control moves to future blocks

Federated peg

A known threshold of keys signs withdrawals

Drivechain peg

Future mainchain blocks accumulate approval
for a withdrawal hash

BIP 300 removes fixed keyholders. It gives withdrawal control to whoever builds enough future block templates.

BIP 300

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- Bitcoin nodes enforcing BIP 300 track this state
- They do not validate the sidechain's transactions or ledger

Six messages

Sidechain slots

M1 proposes
M2 acknowledges

Withdrawal approval

M3 proposes a bundle
M4 changes its score

Peg movement

M5 deposits
M6 withdraws

Treasury output

Current treasury UTXO

`OP_DRIVECHAIN <1-byte slot> OP_TRUE`

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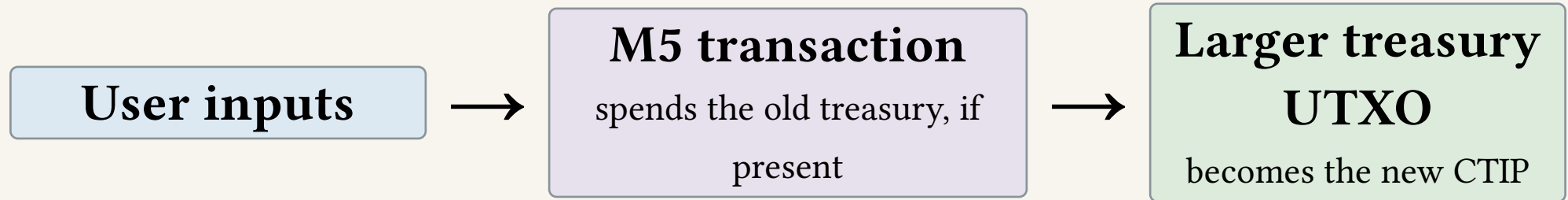
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- Every spend recreates one treasury UTXO for the slot

Deposit path



The sidechain observes the deposit and applies its own rules for crediting the recipient.

Withdrawal path

1. Sidechain users request peg-outs
- 2.
- 3.
- 4.
- 5.
- 6.

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6. M6 pays users and recreates the treasury UTXO with the remaining balance

Vote arithmetic

Selected

+1

Competing

-1

No M4

0

Alarm

-1 to all

- Required score: **13,150**

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- Candidate lifetime: **26,300 blocks**

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- Fastest approval is about 91 days; the full window is about 183 days

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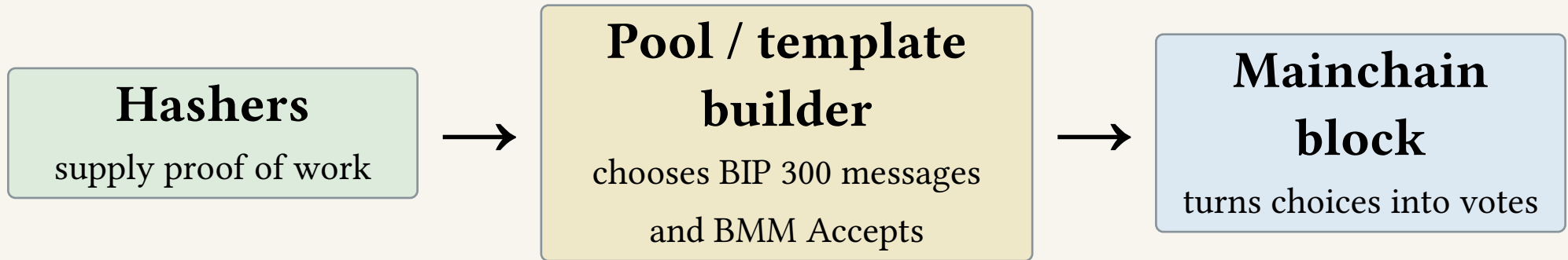
0

Alarm

-1 to all

- Required score: **13,150**
- Candidate lifetime: **26,300 blocks**
- Fastest approval is about 91 days; the full window is about 183 days
- Downvotes increase the number of positive blocks required

Block templates cast the votes



Under ordinary pooled mining, the pool votes. An ASIC owner's fraction of hashrate is not independently represented on-chain.

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- Monitoring can expose a bad bundle
 - Detection does not cryptographically stop it

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- BIP 301 supplies the proposed block-production market

BIP 301

Classical and blind merged mining

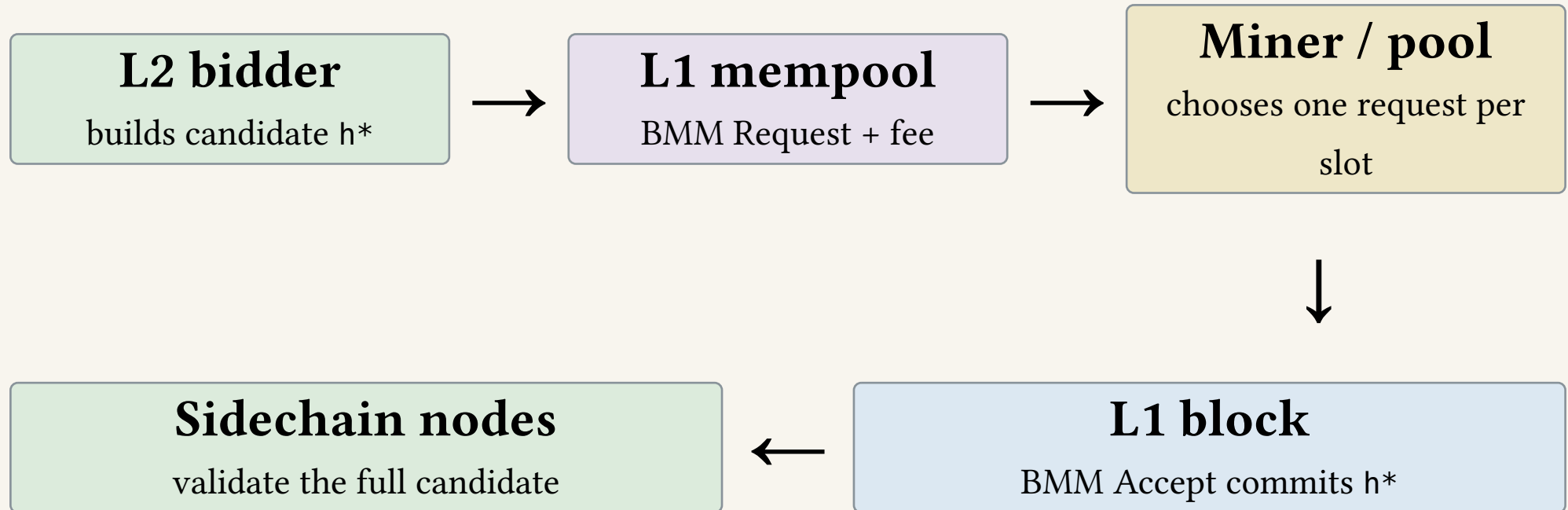
Classical merged mining

Miner runs the other chain's node, validates its block, and receives its native coin

Blind merged mining

Sidechain user builds the block and bids an L1 fee for the miner's commitment

BMM actors



One BMM round

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5. Sidechain nodes validate the candidate and either connect it or reject it

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- An invalid winning candidate buys an empty round; it does not become valid state
- The miner still receives the L1 fee

BIP 300 and 301 together

BIP 301

Which candidate becomes the next
sidechain block?

BIP 300

When may the mainchain treasury release
funds?

BMM anchors the L2 block sequence. It does not validate or secure the
peg-out bundle.

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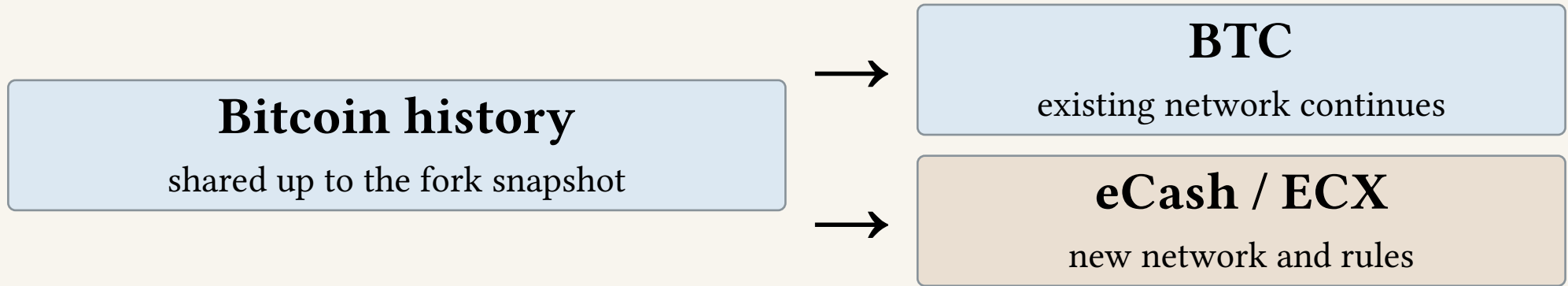
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- An invalid withdrawal accumulates enough score
 - BIP 300 permits the treasury spend; BIP 301 does not prevent it

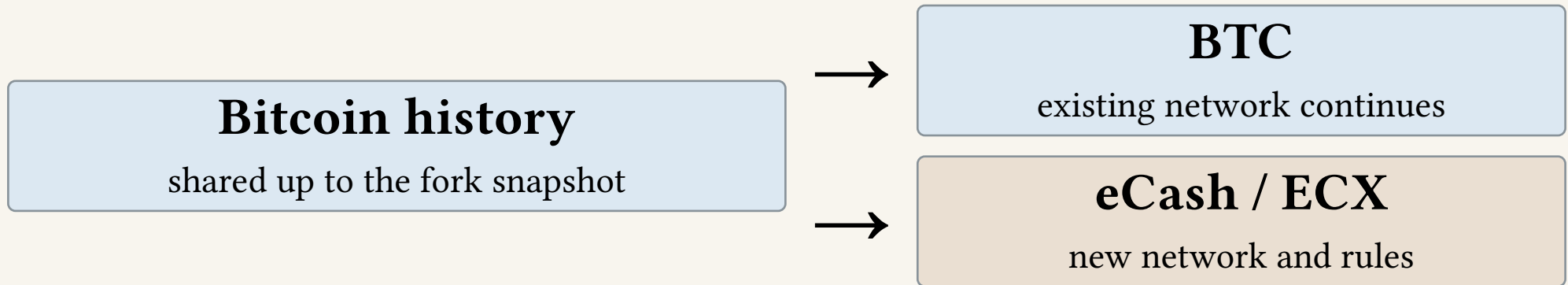
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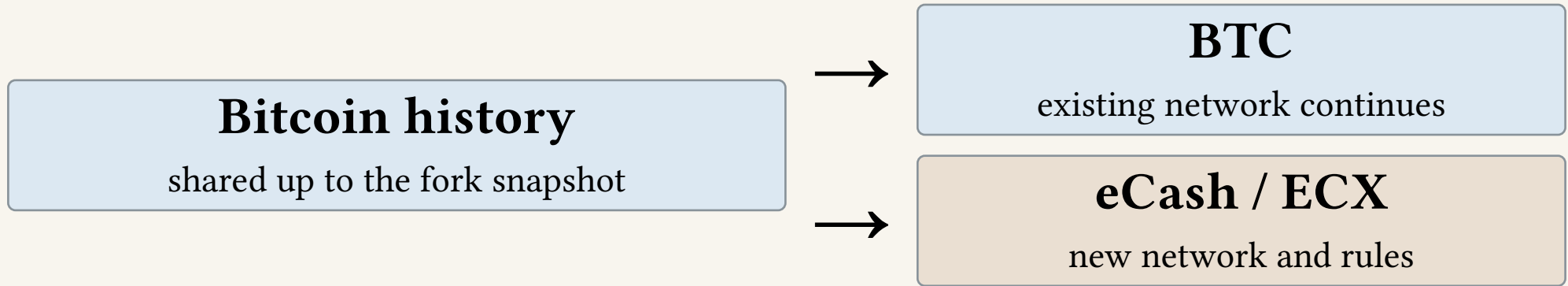
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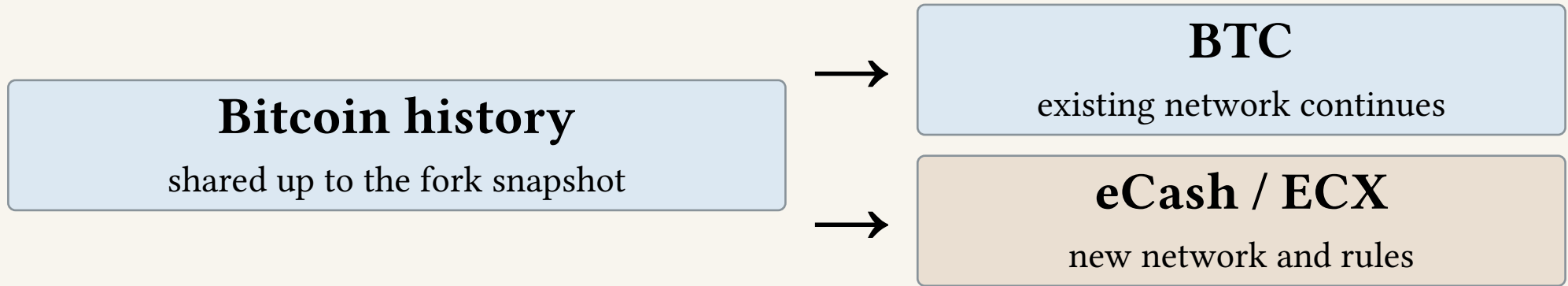
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- Planned SHA-256d chain; Bitcoin itself is unaffected
- This is not Chaumian eCash, Cashu, or the existing XEC network

Fork rules

Fork bootstrap

One-time minimum difficulty

New network magic, seeds, and name

Optional transaction replay-control byte

CUSF-enforced rules

BIP 300 and BIP 301

Project-described 400 kB block cap

The project intends to keep the remaining L1 code close to upstream Bitcoin Core.

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- That reallocation changes only the new chain’s ledger

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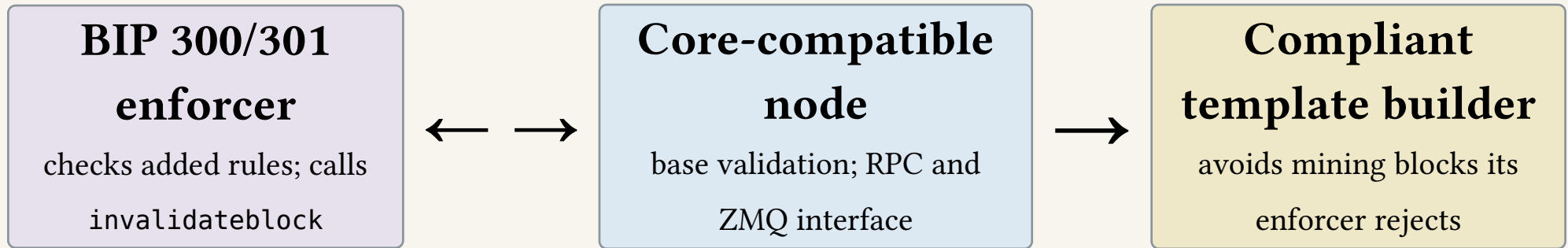
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- eCash plans one period at minimum difficulty
 - Early blocks can arrive quickly with little proof of work
- Normal retargeting resumes after 2,016 eCash blocks
 - Security and block cadence seek a new equilibrium

CUSF stack



CUSF leaves the rules outside Bitcoin Core's source tree. They are still consensus rules for nodes that run the enforcer.

Advertised sidechains

- **Thunder**: high-throughput payments
- **zSide**: Zcash-style privacy
- **BitNames**: identity and DNS
- **BitAssets**: issued assets
- **Truthcoin**: prediction markets
- **CoinShift**: decentralized exchange
- **Photon**: post-quantum signatures

This is the project's current test and launch set; implementation maturity varies.

Assessment

Trust and scope

BIP 300

Removes a fixed federation, but
replaces it with delayed
hashrate custody

BIP 301

Lets miners sell L2
commitments without
validating L2 state

eCash

Runs the experiment on a
separate asset and security
budget

Q&A